

JEE MAIN REVISION MASTER: LAWS OF MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Friction: Static, Kinetic & Inclined Plane Dynamics	6 PYQs	Friction
Moderate	Newton's Laws of Motion & Connected Systems (Pulleys)	5 PYQs	Classical Dynamics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Friction: Static, Kinetic & Inclined Plane Dynamics	$f_{\max} = \mu_s mg \cos \theta$, $a = g(\sin \theta - \mu \cos \theta)$, $\tan \phi = \mu_s$
Newton's Laws of Motion & Connected Systems (Pulleys)	$a = \frac{F_{\text{net}}}{M_{\text{total}}}$, $T_{\text{Atwood}} = \frac{2m_1 m_2 g}{m_1 + m_2}$, $W_{\text{app}} = m(g \pm a)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Friction: Static, Kinetic & Inclined Plane Dynamics (Moderate Yield — 6 PYQs)

Core Principle: Static friction is self-adjusting up to limiting friction $f_L = \mu_s N$. Kinetic friction $f_k = \mu_k N$ (with $\mu_k < \mu_s$). Acceleration down a rough inclined plane: $a = g(\sin \theta - \mu_k \cos \theta)$. Angle of repose $\tan \theta = \mu_s$.

■■ **Caution / JEE Pitfall:** Static friction is NOT always equal to $\mu_s N$; it equals the applied force up to the threshold $\mu_s N$!

■ **Speed Trick:** Time taken to slide down a rough incline compared to smooth incline of same length: $t_{\text{rough}} = n \times t_{\text{smooth}}$ implies $\mu = \tan \theta \left(1 - \frac{1}{n^2}\right)$.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown *i...*" [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "A block of certain mass is placed on a rough floor. The coefficients of static and kinetic friction between the block and the floor are 0.4 ..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "A block of certain mass is placed on a rough inclined plane. The angle between the plane and the horizontal is 30° . The coefficie..." [Ans: N/A]

3.2 Newton's Laws of Motion & Connected Systems (Pulleys) (Moderate Yield — 5 PYQs)

Core Principle: Newton's laws: $\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$, impulse $\vec{J} = \Delta\vec{p} = \vec{F}\Delta t$. Connected systems on Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$, tension $T = \frac{2m_1m_2g}{m_1 + m_2}$. Apparent weight in an elevator accelerating upward is $m(g+a)$; accelerating downward is $m(g-a)$.

■ **Caution / JEE Pitfall:** In a free-falling lift ($a = g$), apparent weight is zero (weightlessness) and normal reaction $N = 0$.

■ **Speed Trick:** For pulley problems, treat the entire string assembly as one 1D system: $a = \frac{\text{(Net Unbalanced Pulling Force)}}{\text{Total Inertial Mass}}$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string...." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 6 m/s^2 . Wha..." **[Ans: N/A]**
- **[2019 Shift] Q#4 (MCQ):** "An object with mass 5kg is acted upon by a force, $\vec{F} = (-3\hat{i} - 4\hat{j})\text{ N}$. If its initial velocity at $t=0$ is $\vec{v} = (3\hat{i} + 12\hat{j})\text{ m/s}$..." **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string.

- (A) $\sqrt{2}Mg$
- (B) $\sqrt{3}Mg$
- (C) Mg
- (D) $3Mg$

Correct Answer: None

Solution Analysis: Let $T_{(1)}$ be the tension in the horizontal string, $T_{(2)}$ be the tension in the slanted string, and $T_{(3)}$ be the tension in the vertical string. The mass M is at rest, so it is in equilibrium. The vertical string supports the weight of the mass. Hence, $T_{(3)} = Mg$. Now consider the knot O . Three tensions act on it: $T_{(1)}$ horizontally, $T_{(2)}$ along the slanted string, and $T_{(3)}$ vertically downward. Since the knot is also in equilibrium, the forces in the vertical direction must balance. $T_{(2)}\sin 45^\circ = T_{(3)} = Mg$. The forces in the horizontal direction must also balance. $T_{(1)} = T_{(2)}\cos 45^\circ = Mg$.

Benchmark Problem 2: KCET 2025 (Q#3)

A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown in the figure. To achieve equilibrium, the co-efficient of static friction between the block on the floor with the floor itself is

- (A) $\mu = \cot\theta$
- (B) $\mu = \sin\theta$
- (C) $\mu = \tan\theta$
- (D) $\mu = \cos\theta$

Correct Answer: None

Solution Analysis: A wooden block of mass M rests on a rough floor, while another block of the same mass is suspended from a point O by strings. To maintain equilibrium, we need to determine the coefficient of static friction between the block on the floor and the floor itself. From the figure and given conditions, we can use the following equations for equilibrium:
The vertical component of tension: $T \sin\theta = Mg$...(i) This equation accounts for the vertical force balance, where T is the tension in the string, and θ is the angle made by the string with the vertical. The horizontal component of